№ T11.

$H_0: \xi \sim p_0(x) = 1\{(0;1)\}$

$H_1: \xi \sim p_1(x) = \frac{e}{e-1} \cdot e^{-x}\{(0;1)\}$

ⓐ $(n=1)$

$$l = \frac{L_1}{L_0} = \frac{\frac{e}{e-1} \cdot e^{-x}}{1} \geq C = const$$

$e^{-x} \geq B \Rightarrow x \leq A$ усл. G

$$P(x \leq A | H_0) = \alpha = \int_0^A 1 \cdot dx = A \Rightarrow G: x \leq \alpha$$

$$\Rightarrow \alpha_1 = \alpha$$

$$W = P(x \leq A | H_1) = \int_0^\alpha \frac{e}{e-1} \cdot e^{-x} dx = \frac{e}{e-1}(1 - e^{-\alpha})$$

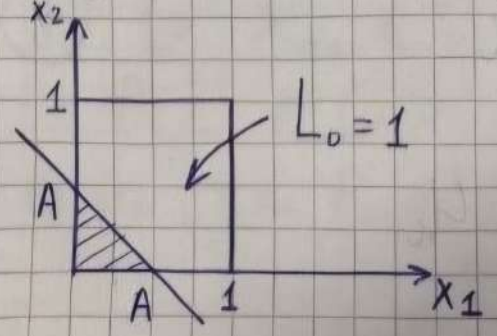
$$\alpha_2 = 1 - W = 1 - \frac{e}{e-1}(1 - e^{-\alpha})$$

ⓑ $$l = \frac{L_1}{L_0} = \frac{\left(\frac{e}{e-1}\right)^2 \cdot e^{-x_1} \cdot e^{-x_2}}{1 \cdot 1} \geq C$$

$e^{-(x_1+x_2)} \geq B \Rightarrow x_1 + x_2 \leq A$ - уст. G.

$P(x_1 + x_2 \leq A | H_0) = \alpha$

(при малых α)



$$\iint_{x_1+x_2 \leq A} 1 \cdot dx_1 \cdot dx_2 = \frac{A^2}{2} = \alpha$$

$$\Rightarrow A = \sqrt{2\alpha}$$

$G: x_1 + x_2 \leq \sqrt{2\alpha}$

$\alpha_1 = \alpha$

$$W = P(X_1 + X_2 \le A \mid H_1) = \iint\limits_{X_1+X_2 \le A} \left(\frac{e}{e-1}\right)^2 e^{-X_1-X_2}\, dX_1\, dX_2 =$$

$$= \left(\frac{e}{e-1}\right)^2 \int_0^A e^{-X_1}\, dX_1 \int_0^{A-X_1} e^{-X_2}\, dX_2 = \left(\frac{e}{e-1}\right)^2 \cdot \int_0^A e^{-X_1}\left(1 - e^{-A+X_1}\right) dX_1 =$$

$$= \frac{-Ae^2 e^{-A}}{(e-1)^2} + \left(\frac{e}{e-1}\right)^2 \cdot \left(1 - e^{-A}\right) = \left(\frac{e}{e-1}\right)^2 \cdot \left(1 - e^{-A} - Ae^{-A}\right)$$

(c) Наиболее мощный асимптотич. критерий.

$$l = \frac{L_1}{L_0} = \prod_{i=1}^{n} \frac{P_1(X_i)}{P_0(X_i)} \geq C$$

$$G: \ln l = \sum_{i=1}^{n} \underbrace{\ln \frac{P_1(X_i)}{P_0(X_i)}}_{\eta_i} \geq \ln C$$

$$\frac{\sum \eta_i - nM\eta_1}{\sqrt{n \cdot D\eta_1}} \sim N(0,1)$$

одинак. распр.

$$P(\ln l \geq \ln C \mid H_0) = \alpha$$

$$\eta = \ln\left(\frac{e}{e-1} \cdot e^{-x}\right) = \ln \frac{e}{e-1} - x$$

$$\ln l = \sum_{i=1}^{n} \ln \frac{e}{e-1} - \sum_{i=1}^{n} X_i \geq \ln C$$

$$G: \sum X_i \le A$$

$$P\left(\frac{\sum X_i - nMX}{\sqrt{nDX}} \le \frac{A - nMX}{\sqrt{nDX}} \,\middle|\, H_0\right) = \alpha$$

для норм. распр.: $\dfrac{A - n \cdot \frac{1}{2}}{\sqrt{n \cdot \frac{1}{12}}} = U_\alpha$

$$A = \frac{n}{2} + U_\alpha \sqrt{\frac{n}{12}}$$

$G: \sum X_i \le n \cdot \frac{1}{2} + U_\alpha \sqrt{\frac{n}{12}}$

$\alpha_1 = \alpha$

$$W = P\left(\sum X_i \le A \mid H_1\right) = P\left(\frac{\sum X_i - nMX}{\sqrt{nDX}} \le \frac{A - nMX}{\sqrt{nDX}} \,\middle|\, H_1\right) =$$

$$MX = \int_0^1 x \frac{e}{e-1} e^{-x}\, dx = \frac{e}{e-1}\left(-xe^{-x}\Big|_0^1 - \int_0^1 e^{-x} dx\right) =$$

$$= \frac{e}{e-1}\left(-e^{-1} + (1 - e^{-1})\right) = \frac{e-2}{e-1}$$

$$MX^2 = \int_0^1 \frac{x^2 \cdot e^{1-x}}{e-1}\, dx = \frac{e}{e-1}\left(-\int_0^1 x^2 de^{-x}\right) =$$

$$= \frac{e}{e-1}\left(-x^2 \cdot e^{-x}\Big|_0^1 + \int_0^1 2xe^{-x} dx\right) = \frac{e}{e-1}\left(-e^{-1} - 2e^{-1} + 2 - 2e^{-1}\right) =$$

$$= \frac{2e-5}{e-1}$$

$$\Rightarrow DX = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$= \int_{-\infty}^{B} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}}\, dx, \text{ где}$$

$$B = \frac{\frac{n}{2} + U_\alpha \sqrt{\frac{n}{12}} - n\frac{e-2}{e-1}}{\sqrt{n\frac{e^2-3e+1}{(e-1)^2}}}$$

$\alpha_2 = 1 - W$

$$B \xrightarrow[n \to \infty]{} +\infty$$

т.о. критерий явл. сост.

(d) $G: X_{min} \le C$ $\qquad H_0: \xi \sim R(0,1)$

$P(\vec{X}_n \in G \mid H_0) = \alpha$

если $\xi \sim F_0(x)$, то $\xi_{min} \sim 1 - (1 - F_0(x))^n$

где $\xi_{\overline{1,n}}$ – незав.

$$P(X_{min} \le C) = 1 - (1 - F(C))^n = \alpha$$

$$\alpha - 1 = -(1-C)^n \Rightarrow C = 1 - \sqrt[n]{1-\alpha}$$

$$W \overset{\alpha = \alpha_1}{=} P(\vec{x_n} \in G \mid H_1) = P(X_{min} \le C \mid H_1) = 1 - (1 - F_1(C))^n$$

где $F_1 = \int_0^x \frac{e}{e-1} \cdot e^{-t}\, dt = \frac{e}{e-1}(1 - e^{-x})$

$$W = 1 - \left(1 - \frac{e}{e-1}\left(1 - e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n =$$

$$= 1 - \left(1 - \frac{e}{e-1} + \frac{e^{\sqrt[n]{1-\alpha}}}{e} \cdot \frac{e}{e-1}\right)^n \xrightarrow[n \to \infty]{} 1$$

Рассмотрим $e^{\sqrt[n]{1-\alpha}} = \exp\left(e^{\frac{1}{n} \cdot \ln(1-\alpha)}\right) \overset{\text{ф. Тейлора}}{=}$

$$= \exp\left(1 + \frac{1}{n}\ln(1-\alpha) + O\left(\frac{1}{n}\right)\right) \underset{\text{ф.Т.}}{\approx} e\left(1 + \frac{1}{n}\ln(1-\alpha) + O\left(\frac{1}{n}\right)\right)$$

$$\Rightarrow W = 1 - \left(1 + \frac{e + \frac{e}{n}\ln(1-\alpha) + O\left(\frac{1}{n}\right)}{e-1} - \frac{e}{e-1}\right)^n =$$

$$= 1 - \left(1 + \frac{e\ln(1-\alpha)}{e-1} \cdot \frac{1}{n} + \underbrace{O\left(\frac{1}{n}\right)}_{\to 0}\right)^n \longrightarrow \exp\left(\frac{e}{e-1}\ln(1-\alpha)\right) =$$

$$= 1 - (1-\alpha)^{\frac{e}{e-1}} \ne 1, \text{ т.к. } (1-\alpha)^{\frac{e}{e-1}} \ne 0$$

Критерий не явл. состоят.